

Exact results for Kerr black holes: shadow areas and perimeters, capture of slow particles, and escape from near extremal horizons

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Abstract

We give exact results for several observables of Kerr black holes that have so far been computed numerically, by fits, or only at isolated parameter values. (i) The area of the shadow seen by an equatorial observer, for every spin, in complete elliptic integrals, together with its small-spin series and its near-extremal expansion $\mathcal{A} = 16\pi + 15\sqrt{3} + (3\pi/\sqrt{2})\sqrt{1-a} + \dots$; the shadow area of an extremal black hole at every inclination; and a single short integral for arbitrary spin and inclination. (ii) The perimeter of the shadow, equal to $18\sqrt{3}M$ for a maximally spinning hole seen edge-on. (iii) The capture cross-section of a Kerr black hole for slow particles arriving perpendicular to the spin, $\sigma = [7\pi + \pi\sqrt{1-a^2} + 16\sqrt{1+a}E(2a/(1+a))]M^2/v^2$, the corresponding angular-momentum accretion in closed form, and the isotropically averaged cross-section and spin-down at maximal spin in elementary closed form. (iv) Escape probabilities for photons emitted isotropically just outside the horizon of an extremal Kerr–Newman black hole at any latitude, $P = \frac{1}{2} - \arcsin(k)/(2\pi) - k/4$, and from the innermost stable circular orbit at any charge, together with escaping energy fractions and escape probabilities for massive particles. Known special cases are recovered and credited. All results were checked against independent high-precision numerical computations.

1 Introduction

The Kerr metric is the unique stationary, asymptotically flat vacuum black hole of general relativity, and many of its observable properties reduce to integrals over the critical (marginally trapped) geodesics. For the shadow—the region of the observer’s sky bounded by the image of the unstable spherical photon orbits [1, 2]—the boundary curve is known in closed parametric form, but derived quantities such as its *area* and *perimeter* have been computed numerically or by interpolation; for instance, Cunha, Herdeiro and Radu remark that they “were not able to find an exact expression for the Kerr shadow areal radius in the generic case” [3]. Likewise the capture cross-section of a Kerr black hole for slow (marginally bound) particles, relevant for black holes moving through dark matter, has been obtained numerically or by fits [10, 11, 13, 14], and the probability that light or matter escapes from just outside the horizon of an extremal black hole has been computed in closed form only at isolated points [18, 19].

In this paper we give exact formulas for these quantities:

- the area of the Kerr shadow seen edge-on, for every spin, in complete elliptic integrals, with its small-spin and near-extremal expansions (Section 3);
- the shadow area of an extremal Kerr black hole at every inclination, and a single-integral representation for arbitrary spin and inclination (Sections 4–5);
- the perimeter of the shadow (Section 6);

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- the edge-on capture cross-section for slow particles, the associated angular-momentum accretion, and the isotropically averaged cross-section and spin-down at maximal spin (Section 7);
- escape probabilities and escaping energy fractions for photons and massive particles emitted isotropically near the horizon of an extremal Kerr(–Newman) black hole, at any latitude and for emitters at rest or on the innermost stable circular orbit (Section 8).

Results that were known before are identified as such in the text, with references. Units are $G = c = M = 1$ throughout; $a = J/M^2 \in [0, 1]$.

2 Shadow boundary

For a distant observer at inclination ϑ to the spin axis ($s = \sin \vartheta$, $c = \cos \vartheta$), the shadow boundary consists of the images of the spherical photon orbits of radius r , with impact parameters [1, 2]

$$\xi(r) = \frac{r^2(3-r) - a^2(r+1)}{a(r-1)}, \quad \eta(r) = \frac{r^3(4a^2 - r(r-3)^2)}{a^2(r-1)^2}, \quad (2.1)$$

and celestial coordinates $\alpha = -\xi/s$, $\beta = \pm\sqrt{\eta + a^2c^2 - \xi^2c^2/s^2}$. The area is $\mathcal{A} = 2 \int \beta |d\alpha|$ over the range of r where $\beta^2 \geq 0$. A key simplification is

$$\frac{d\xi}{dr} = -\frac{2[(r-1)^3 + 1 - a^2]}{a(r-1)^2}. \quad (2.2)$$

The prograde and retrograde circular photon orbits are $r_p = 2[1 + \cos(\frac{2}{3} \arccos(-a))]$, $r_r = 2[1 + \cos(\frac{2}{3} \arccos a)]$; the third root of $r^3 - 6r^2 + 9r - 4a^2$ is $r_0 = 6 - r_p - r_r \in (0, 1)$.

3 Edge-on shadow area for every spin

Result 1. For $\vartheta = \pi/2$ and $0 < a < 1$,

$$\mathcal{A}(a) = 3 \int_{r_p}^{r_r} \frac{(r-1)(5r-2) + 2(1-a^2)/(r-1)}{\sqrt{r(4a^2 - r(r-3)^2)}} dr. \quad (3.1)$$

In terms of complete elliptic integrals $K = K(m)$, $E = E(m)$, $\Pi(n) = \Pi(n|m)$ (parameter convention), with

$$m = \frac{(r_r - r_p)r_0}{(r_r - r_0)r_p}, \quad n = \frac{r_r - r_p}{r_r - r_0}, \quad n' = n \frac{r_0 - 1}{r_p - 1}, \quad g = \frac{2}{\sqrt{(r_r - r_0)r_p}},$$

$$V = \frac{nE + (m-n)K + (2nm + 2n - n^2 - 3m)\Pi(n)}{2(n-1)(m-n)},$$

$$I_0 = gK, \quad I_1 = g[r_0K + (r_p - r_0)\Pi(n)], \quad I_2 = g[r_0^2K + 2r_0(r_p - r_0)\Pi(n) + (r_p - r_0)^2V],$$

$$J = g\left[\frac{K}{r_0 - 1} + \frac{r_0 - r_p}{(r_p - 1)(r_0 - 1)}\Pi(n')\right],$$

$$\mathcal{A}(a) = 6I_0 - 21I_1 + 15I_2 + 6(1 - a^2)J. \quad (3.2)$$

Derivation. Using (2.2), the area integrand is $(4/a^2)r\sqrt{W}[1 + (1 - a^2)/(r - 1)^3]$ with $W = r(4a^2 - r(r - 3)^2)$. Subtracting an exact derivative $\frac{d}{dr}[S(r)\sqrt{W}]$ that vanishes at both endpoints (found by Hermite reduction with SYMPY) leaves (3.1); remarkably the polynomial part of the reduced integrand does not depend on a . The substitution $\text{sn}^2 u = (r_r - r_0)(r - r_p)/[(r_r - r_p)(r - r_0)]$ gives (3.2).

Expansions. The small-spin expansion has rational coefficients (identified from 250-digit evaluations of (3.2); all denominators are of the form $2^p 3^q$):

$$\frac{\mathcal{A}}{27\pi} = 1 - \frac{a^2}{18} - \frac{a^4}{72} - \frac{2059 a^6}{314928} - \frac{88175 a^8}{22674816} - \frac{355999 a^{10}}{136048896} - \dots \quad (3.3)$$

The a^2 term follows from the known order- a^2 elliptical shape of the shadow [5, 6] and is stated explicitly for the area in [8]; the a^4 term is contained in the perturbative results of Kobialko and Gal'tsov [7]. The higher coefficients appear to be new. At maximal spin $\mathcal{A}(1) = 16\pi + 15\sqrt{3}$, a value obtained previously in [3, 4]. Near extremality, with $\epsilon = 1 - a$,

$$\mathcal{A} = 16\pi + 15\sqrt{3} + \frac{3\pi}{\sqrt{2}} \epsilon^{1/2} + \epsilon \left[\frac{8}{\sqrt{3}} \ln \frac{27}{2\epsilon} - 6\sqrt{3} \right] + o(\epsilon). \quad (3.4)$$

The $\epsilon^{1/2}$ term follows from matched asymptotics: as $a \rightarrow 1$ the roots r_0, r_p merge at $r = 1$, $r_{p,0} \simeq 1 \pm \delta$ with $\delta = \sqrt{8\epsilon/3}$; in the inner region the pole term $6(1 - a^2)/(r - 1) \simeq 12\epsilon/(r - 1)$ dominates, and $\int_1^\infty dx/(x\sqrt{x^2 - 1}) = \pi/2$ gives $(12\epsilon/\sqrt{3}\delta)(\pi/2) = 3\pi\epsilon^{1/2}/\sqrt{2}$. The ϵ and $\epsilon \ln \epsilon$ coefficients were identified numerically (evaluating (3.2) with 60–150 digits for ϵ between 10^{-12} and 10^{-80} , followed by an integer-relation search). The square-root onset means that the edge-on shadow area changes rapidly as the spin drops below its maximum.

4 Extremal Kerr at every inclination

For $a = 1$, $\xi = 1 + 2r - r^2$ and $\eta = r^3(4 - r)$. The quartic that defines β^2 factorises:

$$s^2 \left[r^3(4 - r) + c^2 - \frac{c^2}{s^2}(r^2 - 2r - 1)^2 \right] = -(r^2 - 2(1 + s)r - c^2)(r^2 - 2(1 - s)r - c^2), \quad (4.1)$$

with the elementary roots $r_A = 1 - s - \sqrt{2 - 2s} < r_B = 1 + s - \sqrt{2 + 2s} < r_C = 1 - s + \sqrt{2 - 2s} < r_D = 1 + s + \sqrt{2 + 2s}$. (Geometrically, Gralla and Lupsasca showed that the extremal critical curve is the convex hull of a Cartesian oval, completed by the straight “NHEK line” when $\sin \vartheta > \sqrt{3} - 1$ [6]; the factorisation above is the corresponding statement in terms of the orbit radius.)

Result 2. With $\mathcal{R}(r) = (r - r_A)(r - r_B)(r - r_C)(r_D - r)$ and $\vartheta_c = \arcsin(\sqrt{3} - 1) \approx 47.0586^\circ$,

$$\mathcal{A}_{\text{ext}}(\vartheta) = 4 \int_{r_{10}}^{r_D} \frac{3r^2 - \cos^2 \vartheta}{\sqrt{\mathcal{R}(r)}} dr + B(\vartheta), \quad (4.2)$$

where $r_{10} = r_C$, $B = 0$ for $\vartheta \leq \vartheta_c$, and $r_{10} = 1$, $B = (2 + s^2)\sqrt{3s^2 + s^2c^2 - 4c^2}/s^2$ for $\vartheta > \vartheta_c$ (the NHEK-line contribution). For $\vartheta \leq \vartheta_c$, with $(e_1, e_2, e_3, e_4) = (r_D, r_C, r_B, r_A)$, $m = \frac{(e_1 - e_2)(e_3 - e_4)}{(e_1 - e_3)(e_2 - e_4)}$, $n = \frac{e_1 - e_2}{e_1 - e_3}$, $g = 2/\sqrt{(e_1 - e_3)(e_2 - e_4)}$, V as in Result 1,

$$\mathcal{A}_{\text{ext}} = 4(3I_2 - c^2 I_0), \quad I_0 = gK, \quad I_2 = g[e_3^2 K + 2e_3(e_2 - e_3)\Pi(n) + (e_2 - e_3)^2 V]. \quad (4.3)$$

The reduction uses, with $x = r - 1$ and $P(x) \equiv \mathcal{R}(1 + x)$, the identity $x\sqrt{P} = s^2(3x^2 + 6x + s^2 + 2)/\sqrt{P} + \frac{d}{dx}[(x^2 - s^2 - 2)\sqrt{P}/4]$. The limits are the known values $\pi(12 + 8\sqrt{2})$ at $\vartheta = 0$ [4] and $16\pi + 15\sqrt{3}$ at $\vartheta = \pi/2$. Near the axis,

$$\mathcal{A}_{\text{ext}} = \pi(12 + 8\sqrt{2}) + \sqrt{2}\pi \left[\frac{s^2}{2} + \frac{s^4}{128} - \frac{7s^6}{2048} - \frac{1735s^8}{524288} - \dots \right] \quad (4.4)$$

(coefficients identified by integer-relation search). For $a = 1 - \epsilon$ the area grows as $\mathcal{A}_{\text{ext}}(\vartheta) + c(\vartheta)\sqrt{\epsilon}$ with

$$c(\vartheta) = \frac{\pi(3 - 6\cos^2 \vartheta - \cos^4 \vartheta)}{\sqrt{2} \sin^3 \vartheta} \quad (\vartheta > \vartheta_c), \quad c(\vartheta) = 0 \quad (\vartheta < \vartheta_c), \quad (4.5)$$

derived as in Section 3 from the master formula below; equivalently $c = \pi\beta_m^2/(\sqrt{2} \sin \vartheta)$ with β_m the half-length of the NHEK line. Sample values: $\mathcal{A}_{\text{ext}}(17^\circ) = 73.43231016343180$, $\mathcal{A}_{\text{ext}}(45^\circ) = 74.35818308197668$, $\mathcal{A}_{\text{ext}}(60^\circ) = 75.08875773057813$.

5 Single-integral formula for any spin and inclination

For general (a, ϑ) the boundary involves a sextic, and no reduction to complete elliptic integrals is expected (the corresponding curve has genus two). Nevertheless the area reduces to one short integral.

Result 3. *With $u = a^2 \cos^2 \vartheta$,*

$$N(r) = -r^6 + 6r^5 - (9 + 2u)r^4 + 4a^2r^3 + 6ur^2 - 4a^2ur - u^2(r-1)^2$$

(so that $N = \beta^2 a^2 (r-1)^2 \sin^2 \vartheta$) and $r_1 < r_2$ the roots of N with $r > 1$ that bound the shadow,

$$\mathcal{A}(a, \vartheta) = \frac{1}{u-1} \int_{r_1}^{r_2} \left[2(u^2 - 5a^2u + 6u + 3a^2 - 3) - 2(2u^2 + a^2u - 3u - 3a^2 + 3)r + 2(u^2 - 6u - 3)r^2 + 4(u+3)r^3 + 2(u-3)r^4 - \frac{2(a^2-1)(u^2+6u-3)}{r-1} \right] \frac{dr}{\sqrt{N(r)}}. \quad (5.1)$$

At $\vartheta = \pi/2$ this reduces to (3.1). The integrand has only inverse-square-root endpoint singularities, so tanh–sinh quadrature converges to 15–30 digits at negligible cost.

6 Perimeter of the shadow

Result 4. *Along the critical curve the arc-length element simplifies to $(dl/dr)^2 = 16r(r^2 - a^2c^2)[(r-1)^3 + 1 - a^2]^2 / ((r-1)^4 N(r))$, so for any (a, ϑ)*

$$P(a, \vartheta) = 8 \int_{r_1}^{r_2} \frac{[(r-1)^3 + 1 - a^2] \sqrt{r(r^2 - a^2 \cos^2 \vartheta)}}{(r-1)^2 \sqrt{N(r)}} dr. \quad (6.1)$$

Edge-on only a cubic remains, $P(a) = 8 \int_{r_p}^{r_r} [(r-1) + (1-a^2)/(r-1)^2] dr / \sqrt{4a^2 - r(r-3)^2}$, and in closed form, with $m = (r_r - r_p)/(r_r - r_0)$, $n = (r_r - r_p)/(r_r - 1)$ and V as above,

$$P(a) = \frac{16}{\sqrt{r_r - r_0}} \left[r_0 K + (r_r - r_0) E - K + \frac{(1-a^2)V}{(r_r - 1)^2} \right]. \quad (6.2)$$

At maximal spin edge-on, $dl = 4 dr / \sqrt{4-r}$ on $1 \leq r \leq 4$, so the curved part (both halves) has length $16\sqrt{3}$ and the NHEK line $2\sqrt{3}$:

$$P(1, \pi/2) = 18\sqrt{3} \approx 31.17691. \quad (6.3)$$

Wei, Liu and Mann [9] reported the length $16\sqrt{3}$ at $a = 1$; this is the length of the curved part only. Including the straight NHEK segment, which is part of the critical curve, gives $18\sqrt{3}$, continuous with their value 31.2636 at $a = 0.99$. For small spin (coefficients identified at 60 digits),

$$\frac{P}{6\sqrt{3}\pi} = 1 - \frac{(1+C)a^2}{36} - \left(\frac{35}{5184} + \frac{35C}{7776} - \frac{23C^2}{15552} \right) a^4 - \dots, \quad C = \cos^2 \vartheta. \quad (6.4)$$

7 Capture of slow particles

For particles arriving from infinity with speed $v \ll 1$ the capture boundary in the plane of specific angular momenta is traced by the marginally bound ($E = 1$) spherical orbits. Writing $r = y^2$,

$$L_z = -\frac{y^4 - 2y^3 + a^2}{a(y-1)}, \quad Q = \frac{y^4(a^2 - (y^2 - 2y)^2)}{a^2(y-1)^2}, \quad 1 + \sqrt{1-a} \leq y \leq 1 + \sqrt{1+a}, \quad (7.1)$$

which is equivalent to the marginally bound Carter constant given by Teo [12]. In particular $L_z^2 + Q = (3y^4 - 4y^3 + a^2)/(y-1)^2$. The impact parameters are $(L_z, \sqrt{Q})/v$ and the cross-section is $\sigma = (M/v)^2 \mathcal{S}$ with $\mathcal{S} = 2 \int \sqrt{Q} |dL_z|$.

Result 5 (Edge-on cross-section). *For particles arriving perpendicular to the spin axis,*

$$\sigma = \frac{M^2}{v^2} \left[7\pi + \pi\sqrt{1-a^2} + 16\sqrt{1+a} E\left(\frac{2a}{1+a}\right) \right], \quad (7.2)$$

which gives $16\pi M^2/v^2$ at $a = 0$ and $(7\pi + 16\sqrt{2})M^2/v^2$ at $a = 1$.

With $u = y - 1$ the square root becomes $\sqrt{a^2 - (u^2 - 1)^2}$, an even quartic; Hermite reduction gives $\mathcal{S} = \int_{\sqrt{1-a}}^{\sqrt{1+a}} (16u^2 + 14u + 2(1-a^2)/u) du / \sqrt{a^2 - (u^2 - 1)^2}$, whose odd terms are elementary. The expansion $\mathcal{S}/(16\pi) = 1 - a^2/16 - 31a^4/2048 - 233a^6/32768 - \dots$ agrees through a^4 with the slow limits of the fitted coefficients of Karydas, Vicente and Bertone [13], and the a^2 term with Mach, Momennia and Sarbach [14].

Result 6 (Angular momentum accreted, edge-on). *The first moment of the capture region, $M_1 = \iint L_z dL_z d\sqrt{Q}$, is*

$$M_1 = -\frac{32\sqrt{1+a} [(a^2 + 2)E - 2(1-a)K] + 35\pi a^2 + 10\pi(1 - \sqrt{1-a^2})}{5a}, \quad (7.3)$$

with $K = K(2a/(1+a))$, $E = E(2a/(1+a))$, so a black hole moving through slow matter in its equatorial plane gains angular momentum per unit mass $dJ/dM = M M_1/\mathcal{S} = -aM(1 + \frac{3}{32}a^2 + \frac{37}{1024}a^4 + \dots)$, and $M_1 \rightarrow -(96\sqrt{2} + 45\pi)/5$ as $a \rightarrow 1$.

The a and a^3 terms are consistent with the slow limits of the fits of [13].

Result 7 (Isotropic flux). *For an isotropic flux of slow particles, the direction-averaged cross-section is $\langle\sigma\rangle = (\pi/2)(M/v)^2 \int_{-1}^1 L_c(\mu)^2 d\mu$, where $\mu = \cos i = L_z/L_c$ is the inclination of the orbital angular momentum, giving*

$$\langle\sigma\rangle = \frac{\pi M^2}{2 v^2} \left| \int_{1+\sqrt{1-a}}^{1+\sqrt{1+a}} \frac{2y^3(3y^4 - 8y^3 + 6y^2 - a^2)}{a(y-1)^2 \sqrt{3y^4 - 4y^3 + a^2}} dy \right|. \quad (7.4)$$

At $a = 1$ (where $3y^4 - 4y^3 + 1 = (y-1)^2(3y^2 + 2y + 1)$ and the boundary includes the straight piece $L_z = 2$ for $\mu \geq \sqrt{2/3}$ [11]) everything is elementary:

$$\langle\sigma\rangle_{a=1} = \pi \frac{M^2}{v^2} \left[\frac{47}{9} + \frac{31\sqrt{2}}{6} + \frac{7\sqrt{6}}{9} - \frac{\sqrt{3}}{54} \ln \frac{1 + \sqrt{2} + \sqrt{6}}{1 + \sqrt{2}} \right] = 45.27564306526540 \frac{M^2}{v^2}, \quad (7.5)$$

i.e. 0.90073 of the Schwarzschild value $16\pi M^2/v^2$ (Young reported the ratio 0.90 [10]). The mean axial angular momentum per unit captured mass is $\langle l_z \rangle = \frac{2}{3} \int \mu L_c^3 d\mu / \int L_c^2 d\mu$, and at $a = 1$

$$\int_{-1}^1 \mu L_c^3 d\mu = -\frac{4876}{189} - \frac{790\sqrt{2}}{63} + \frac{628\sqrt{6}}{189} - \frac{2\sqrt{3}}{81} \ln \frac{1 + \sqrt{2} + \sqrt{6}}{1 + \sqrt{2}}, \quad (7.6)$$

$$\langle l_z \rangle = -0.8193268418025525 M. \quad (7.7)$$

The uniform measure over incoming directions and impact-plane azimuths induces the uniform measure on the direction of the orbital angular momentum, which justifies the averages. The small-spin series $\langle l_z \rangle = -(2a/3 + a^3/15 + \dots)M$ is contained in [14, 15]. Young [10] reported -0.828 for this quantity at $a = 1$; our exact value differs by about 1%. It was confirmed by an independent brute-force computation (-0.819328 , Section 9); we have not been able to consult the full text of [10] to identify the origin of the difference (numerical accuracy or a different averaging convention).

Remark 1 (Any speed, maximal spin). For $a = 1$ and particles of any energy $\mathcal{E} = (1 - v^2)^{-1/2}$, $p^2 = \mathcal{E}^2 - 1$, arriving edge-on, the critical orbits are $L = \mathcal{E}(r + 1) - (r - 1)\sqrt{r(p^2 r + 1)}$, $Q = r^2[-p^2 r^2 + (2p^2 - 1)r + 1 + 2\mathcal{E}\sqrt{r(p^2 r + 1)}]$ for $1 \leq r \leq r_{\max}$, and the substitution $r = 1/(t^2 - p^2)$ reduces the cross-section to an incomplete elliptic integral with the quartic $t^4 + 2\mathcal{E}t^3 - t^2 - 2\mathcal{E}p^2 t - p^4$. As v increases, σv^2 starts from $7\pi + 16\sqrt{2}$ ($v \rightarrow 0$), and $\sigma \rightarrow 16\pi + 15\sqrt{3}$ as $v \rightarrow 1$.

8 Escape from near the horizon of an extremal black hole

Consider the extremal Kerr–Newman family $a^2 + q^2 = 1$, with q the charge (Kerr: $a = 1$) and an emitter at $r \rightarrow 1^+$ at polar angle ϑ , emitting isotropically in its rest frame. For extremal black holes a finite fraction of the emission escapes to infinity even in this limit [16, 18, 19].

Result 8 (Emitter at rest). *For an emitter at rest in the zero-angular-momentum frame, let $k = (1 + a^2 \cos^2 \vartheta)/(2a \sin \vartheta)$. The escape probability of photons is*

$$P = \frac{1}{2} - \frac{\arcsin k}{2\pi} - \frac{k}{4} \quad (k < 1), \quad P = 0 \quad (k \geq 1). \quad (8.1)$$

The escape cone is $\{n_\varphi > 0, n_\vartheta^2 < (k^{-2} - 1)n_\varphi^2, (n_\varphi > k \text{ or } n_r > 0)\}$ in terms of the emitter-frame unit direction $(n_r, n_\vartheta, n_\varphi)$.

On the equator of extremal Kerr ($k = 1/2$) this gives $7/24$, the value of Yan, Guo and Chen [18] (numerically 0.2916 in [16]). On the equator of extremal Kerr–Newman ($k = 1/(2a)$) it reproduces the numerical values of Ogasawara et al. [16] (Table II there: 0.267, 0.194, 0.0911 for $a = 0.9, 0.7, 0.55$; formula: 0.26736, 0.19480, 0.09112) and their threshold $a > 1/2$. For Kerr, $P > 0$ iff $\sin \vartheta > \sqrt{3} - 1$, the latitude range singled out by Ogasawara and Igata [20]; the edge $n_\vartheta^2 = (k^{-2} - 1)n_\varphi^2$ corresponds to the near-horizon limit of the exact escape-cone boundary of Baines and Visser [21]. (The $7/24$ value refers to the zero-angular-momentum frame; cone boundaries expressed in other tetrads give different solid angles.)

Derivation. Put $r = 1 + \varepsilon x$ and $\lambda = \lambda_H - \varepsilon u/a$, $\lambda_H = (1 + a^2)/a$. In the zero-angular-momentum frame $n_\varphi = \Sigma_H/(a \sin \vartheta (u + 2))$, $\Sigma_H = 1 + a^2 \cos^2 \vartheta$. The radial potential is $(r - 1)^2[(r + 1)^2 - \mathcal{C}]$ far from the horizon and $\varepsilon^2[(2x + u)^2 - \mathcal{C}x^2]$ near it, with $\mathcal{C} = \eta + 1/a^2 = (u + 2)^2(n_\vartheta^2 + n_\varphi^2)$; escape requires $\mathcal{C} < 4$, and either outward emission or $u < 0$ (inward photons then bounce at $x = |u|/(2 - \sqrt{\mathcal{C}})$). The solid angle of the resulting cone is $\pi - 2 \arcsin k + \pi(1 - k)$.

Result 9 (Emitter on the ISCO). *For extremal Kerr–Newman with $a \geq 1/\sqrt{2}$ the innermost stable circular orbit lies on the horizon and moves with speed $v = k = 1/(2a)$ relative to the zero-angular-momentum frame. A photon emitter on it escapes with probability*

$$P_{\text{ISCO}} = \frac{1}{2} - \frac{\arcsin k}{2\pi} + \frac{k}{\pi\sqrt{1+k^2}} \arctan \sqrt{\frac{1+k^2}{1-k^2}}, \quad k = \frac{1}{2a}. \quad (8.2)$$

At $a = 1$ this is $5/12 + \arctan \sqrt{5/3}/(\sqrt{5}\pi) = 0.546455$, the value found by Gates, Hadar and Lupsasca [19] (numerically 54.6% in [17]). At $a = 1/\sqrt{2}$ it equals $3/8 + \sqrt{3}/9 = 0.567450$. It follows from boosting the cone of Result 8: in the emitter frame, with t the cosine of the angle to the orbital motion, all directions with $t > 0$ escape, none with $t < -k$, and for $-k < t < 0$ a fraction $\frac{1}{\pi} \arcsin \frac{t+k}{k\sqrt{1-t^2}}$, so $P = \frac{1}{2} + \frac{1}{2\pi} \int_{-k}^0 \arcsin \frac{t+k}{k\sqrt{1-t^2}} dt$, which integrates to (8.2).

Result 10 (Escaping energy fractions). *The fraction of the emitted energy (in the emitter frame) that reaches infinity is, for the emitter at rest on the equator ($a \geq 1/2$),*

$$\mathcal{F}_{\text{rest}} = \frac{4a^2 - 1}{32a} + \frac{\sqrt{4a^2 - 1}}{16} \quad (\text{Kerr: } \frac{3}{32} + \frac{\sqrt{3}}{16} = 0.202003), \quad (8.3)$$

and for the ISCO emitter of extremal Kerr

$$\mathcal{F}_{\text{ISCO}} = \frac{2\sqrt{3}}{9} + \frac{1}{20\pi} + \frac{13 \arctan \sqrt{5/3}}{5\sqrt{15}\pi} = 0.595642. \quad (8.4)$$

These follow by weighting the escape regions with the energy ratio $g = E_\infty/E_{\text{emit}}$, which equals $a n_\varphi$ for the emitter at rest and $a(t + k)/\sqrt{1 - k^2}$ for the ISCO emitter. The ISCO value can also be obtained by integrating the redshift-resolved flux of Gates, Hadar and Lupsasca [19] (their Eq. (102)); we did not find the integrated fraction stated there.

Result 11 (Massive particles). (i) For particles emitted isotropically with local speed β by an emitter at rest on the equator of an extremal Kerr–Newman black hole, escape is impossible for $\beta < 1/\sqrt{1+a^2}$ (Kerr: $1/\sqrt{2}$), and

$$P = \frac{1}{2} \left(1 - \frac{\sqrt{1-\beta^2}}{a\beta} \right) \quad \left(\frac{1}{\sqrt{1+a^2}} \leq \beta \leq \frac{\sqrt{3}}{2} \right); \quad (8.5)$$

for Kerr, $P = 1/8$ exactly at $\beta = 0.8$. (ii) For particles launched isotropically with speed β from the ISCO of extremal Kerr, escape requires $\beta > (3\sqrt{2} - 2)/7 = 0.320377$, and

$$P = \frac{2\beta + 1 - \sqrt{3(1-\beta^2)}}{4\beta} \quad \left(\frac{3\sqrt{2} - 2}{7} \leq \beta \leq \sqrt{\frac{2}{3}} \right), \quad (8.6)$$

with $P = 1/4$ at $\beta = 1/2$ and $P = 1/2$ at $\beta = \sqrt{2/3}$.

These follow from the same near-horizon analysis with massive geodesics: escape requires the conserved energy per unit mass to satisfy $\mathcal{E} \geq 1$, the absence of a turning point in the far zone ($3\mathcal{E}^2 - 1 > Q$ for Kerr, with Q the Carter constant) and the near-zone bounce condition. Above the stated ranges the probabilities are given by one-dimensional integrals.

9 Verification

All results were checked against independent computations (scripts in the code repository):

- closed forms against direct quadrature of the original parametric boundary curves at 30–40 digits (e.g. edge-on area at $a = 0.5$: 83.5610541284615455783534950 from both);
- areas against polygon (shoelace) areas of the boundary curves (agreement $\sim 10^{-7}$ – 10^{-9} , limited by the number of vertices);
- the extremal formula at 60° and 80° against the master formula at $a = 1 - 10^{-12}$ including the $c(\vartheta)\sqrt{\epsilon}$ correction;
- slow-capture cross-sections against brute-force bisection with the exact $E = 1$ radial potential ($a = 0.5$: 49.4258 vs 49.4259; $a = 0.9$: 46.8049 vs 46.8048), and the isotropic $a = 1$ values against direct bisection for $L_c(\mu)$ at 2000 inclinations ($\langle l_z \rangle = -0.819328$ vs -0.819327);
- escape probabilities against ray-by-ray simulations: grids of up to 1.8×10^5 emission directions at $r = 1 + 10^{-7}$, exact constants of motion computed without cancellation, escape decided from the exact radial potential including inward-launched rays (agreement within the grid resolution, $\sim 10^{-3}$; e.g. Kerr equator 0.29172 vs $7/24$, ISCO $a = 0.8$: 0.55890 vs 0.55890, massive $\beta = 0.8$: 0.12502 vs $1/8$).

AI disclosure statement

This work was carried out by the author with substantial assistance from an AI system, Claude (Anthropic), used through the Claude Code software. Under the author’s direction, the AI system proposed the research directions, wrote and ran the computer-algebra and numerical code (Python with SymPy and mpmath), performed the derivations and checks reported here, searched the literature, and drafted the text of this paper. The author directed the work, set the requirements (verification of every result, honest novelty assessment, no fabricated citations) and is responsible for its publication. All results were checked numerically as described in the text; nevertheless, readers should treat the novelty assessment, which is based on literature searches, with appropriate caution.

Code availability

All code used to derive and check the results is available at <https://github.com/JacobGoodchild/findformula2> (directory `code/`, and the verification scripts `papers/paperB/verify_paperB.py` and `papers/paperB/verify_paperB_escape.py`).

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